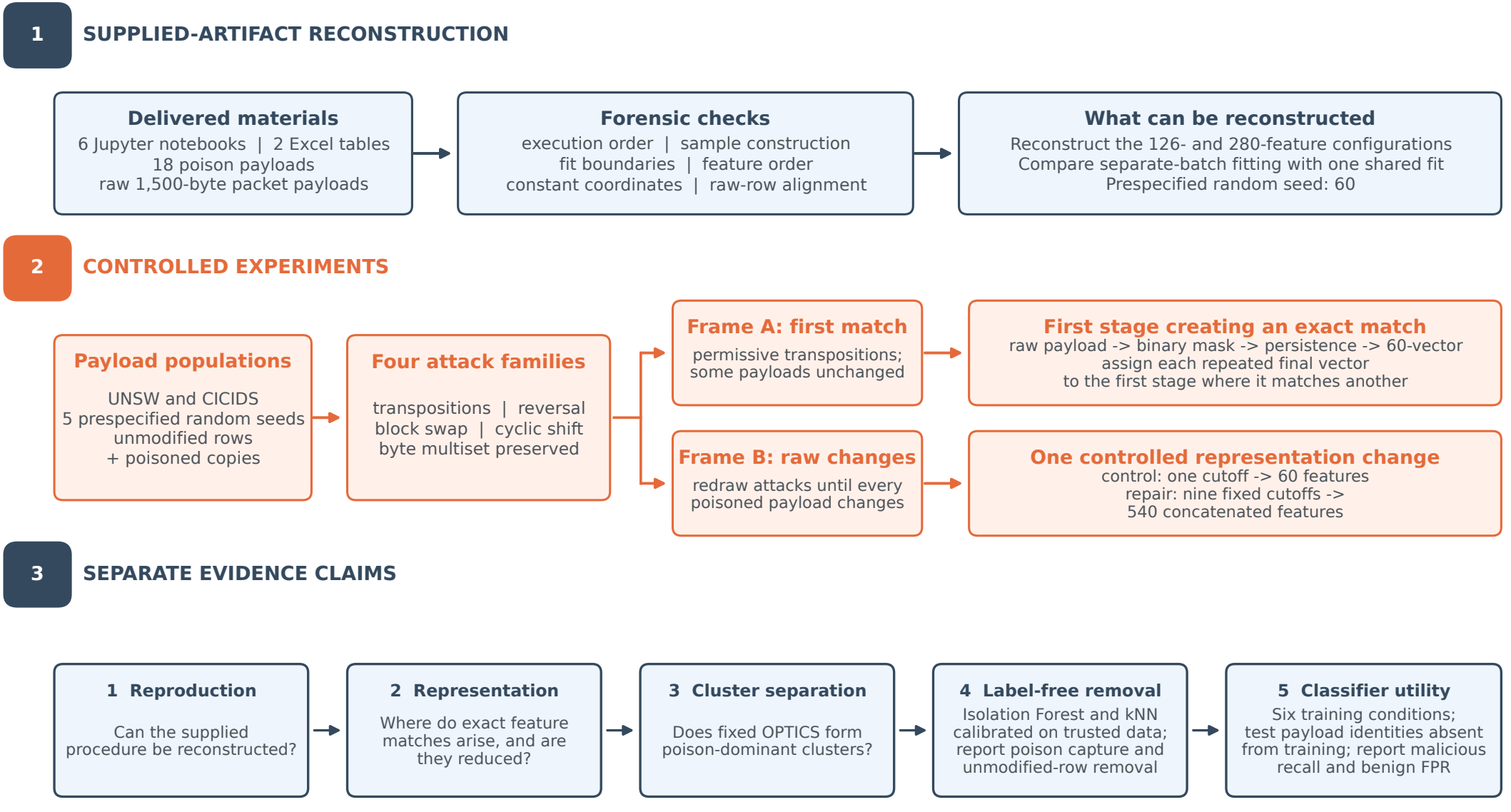
1
SUPPLIED-ARTIFACT RECONSTRUCTION
Delivered materials
6 Jupyter notebooks | 2 Excel tables
18 poison payloads
raw 1,500-byte packet payloads
Forensic checks
execution order | sample construction
fit boundaries | feature order
constant coordinates | raw-row alignment
What can be reconstructed
Reconstruct the 126- and 280-feature configurations
Compare separate-batch fitting with one shared fit
Prespecified random seed: 60
2
CONTROLLED EXPERIMENTS
Payload populations
UNSW and CICIDS
5 prespecified random seeds
unmodified rows
+ poisoned copies
Four attack families
transpositions | reversal
block swap | cyclic shift
byte multiset preserved
Frame A: first match
permissive transpositions;
some payloads unchanged
First stage creating an exact match
raw payload -> binary mask -> persistence -> 60-vector
assign each repeated final vector
to the first stage where it matches another
Frame B: raw changes
redraw attacks until every
poisoned payload changes
One controlled representation change
control: one cutoff -> 60 features
repair: nine fixed cutoffs ->
540 concatenated features
3
SEPARATE EVIDENCE CLAIMS
1 Reproduction
Can the supplied procedure be reconstructed?
2 Representation
Where do exact feature matches arise, and are they reduced?
3 Cluster separation
Does fixed OPTICS form poison-dominant clusters?
4 Label-free removal
Isolation Forest and kNN calibrated on trusted data; report poison capture and unmodified-row removal
5 Classifier utility
Six training conditions; test payload identities absent from training; report malicious recall and benign FPR